

2. Suppose Coca-Cola stock trades at a forward price to earnings ratio of 28, its expected dividend payout ratio is 70 percent, and analysts believe that its required return is 6.50 percent. Calculate Coca-Cola's implied growth rate.

Solution:

4.00 percent

Using Equation 24,

$$28 = \frac{0.70}{0.065 - g}.$$

Solving this equation for g , Coca-Cola's implied growth rate is 4.00 percent. In particular, Coca-Cola's return is expected to be 2.5 percent greater than its dividend growth rate (i.e., $6.50\% - 4.00\%$).

As discussed earlier, the same principles apply to understanding required returns and growth rates on stock indexes.

3. A stock index is trading at a forward price to earnings ratio of 19. If the expected dividend payout ratio on the index is 60 percent, and equity investors expect an index rate of return of 8 percent, calculate the implied constant growth rate for the index.

Solution:

4.84 percent

Using Equation 24:

$$19 = \frac{0.60}{0.08 - g}.$$

Solving for g , we find that investors expect the index's dividend growth rate to be 4.84 percent in the future. Alternatively, we could assume that investors expect a 4.84 percent growth rate and solve for r to calculate 8 percent as the required return. Thus, the index forward price-to-earnings ratio of 19 and an expected dividend payout ratio of 60 percent combine to reflect expectations of 8 percent return and 4.84 percent growth. Specifically, the required return exceeds the implied dividend growth rate by 3.16 percent (i.e., $8\% - 4.84\%$).

An important point from the prior results is that equity prices, whether expressed simply as price or as a price-to-earnings ratio, reflect combined expectations about future returns and growth. Expectations of returns and growth are linked together by the difference between r and g . For example, the Coca-Cola example using a price-to-earnings ratio of 28 describes a situation in which investors must believe that the required return on Coca-Cola stock is 2.5 percent above its growth rate.

4. Suppose Coca-Cola stock trades at a forward price to earnings ratio of 28 and its expected dividend payout ratio is 70 percent. Analysts believe that Coca-Cola stock should earn a 9 percent return and that its dividends will grow by 4.50 percent per year indefinitely. Recommend a course of action for an investor interested in taking a position in Coca-Cola stock.

Solution:

Take a short position in Coca-Cola stock.

If we evaluate the above parameters using Equation 24, we can see that this results in an inequality.